House Price Prediction Using Machine Learning

Objective

The objective of this project was to analyze housing data and build machine learning models to predict house prices based on various features such as area, number of bedrooms, bathrooms, parking spaces, and amenities.

Data Preprocessing

The dataset contained 545 records and 13 attributes, including both numerical and categorical variables. Duplicate rows were checked and removed if present. The target variable (`price`) was separated from the input features, and categorical variables were converted into numerical form using One-Hot Encoding.

Exploratory Data Analysis

A histogram of house prices showed the distribution of property values in the dataset. A correlation heatmap was used to identify relationships between features and the target variable. The analysis revealed that area and bathrooms had the strongest positive correlation with house price, followed by air conditioning, number of stories, and parking availability.

Model Development and Evaluation

Two regression models were trained and compared: Linear Regression and Random Forest Regressor. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score.

Model	MAE	RMSE	R^2 Score
Linear Regressor	779,346	1,061,503	0.6729
Random Forest Regressor	865,137	1,210,614	0.5745

The results indicate that Linear Regression outperformed Random Forest on all evaluation metrics, suggesting that the relationship between the input features and house prices is largely linear.

Conclusion

The project demonstrated that house prices are primarily influenced by property area, number of bathrooms, and modern amenities. Linear Regression achieved an R^2 score of approximately 67%, providing reasonably accurate predictions. The findings suggest that real estate businesses should prioritize larger properties with desirable amenities, as these features are strongly associated with higher market values.

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